

Dynamics I, Mass and Weight Main Note

Force $\Rightarrow$ is a push or pull.

$\Rightarrow$ force is an interaction between objects to explain changes in motion

When force is exerted on an object:

- object can either start or stop moving
- slow down or speed up
- change direction of motion of object

Types of Forces

two types $\Rightarrow$ contact and non contact.

Non contact $\Rightarrow$ do not require objects to be in contact to exist.

Non Contact Forces	Contact Forces
Gravitational force: - pull exerted by earth's gravity on an any object (weight)	Friction - force that tends to oppose or tends to oppose motion between surfaces in contact
Electrostatic force: - attractive (i.e. pull) or repulsive (i.e. push) forces between electric charges. (attractive forces between unlike charges) (repulsive forces between like charges)	Air Resistance - frictional force exerted by air that opposes motion of moving objects.
Magnetic force Attractive (i.e. pull) or repulsive (i.e. push) forces between magnets. (attractive magnetic forces between unlike poles) (repulsive magnetic forces between like poles)	Normal Force - push exerted by a surface on an object pressing on it - push is always perpendicular to the surface
	Tension - pull exerted by a stretched spring, string, rope on an object attached to it

Mass

Definition: Mass is a measure of amount of matter in a body

#PhysicsDefinitions

SI unit: Kg

Mass $\Rightarrow$ property of body $\Rightarrow$ does not change with its location or shape

Mass of body $\Rightarrow$ depends on the number and composition of atoms and molecules that make up the body

Scalar quantity

Weight

Definition: Weight is the gravitational force acting on an object that has mass

#PhysicsDefinitions

SI unit: Newton(N)

Since weight is a force $\Rightarrow$ it is a **Vector quantity** $\rightarrow$ with both magnitude and direction.
Direction of weight $\Rightarrow$ downwards $\Rightarrow$ i.e. towards center of earth

Gravitational Field

Definition: gravitational field is a region in which a mass experiences a force due to gravitational attraction

#PhysicsDefinitions

Example:

- earth with huge mass $\Rightarrow$ has gravitational field surrounding it
 - thus any object within earth's gravitational field $\Rightarrow$ will experience a force exerted by earth on it
 - gravitational force experienced is strongest at surface of earth
 - gets weaker further away due to decreasing gravitational field strength
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Gravitational Field Strength

Weight of any object $\Rightarrow$ depends on strength of gravitational force acting on it.

Eg: objects weight less on the moon than on earth because moon's gravitational field strength is weaker than earth's gravitational field strength

Definition: Gravitational field strength, g is defined as the gravitational force per unit mass placed at one point

#PhysicsDefinitions

Formula for gravitational field strength:

$$g = \frac{W}{m}$$

Where g , is gravitational field strength (N/Kg)

W = weight (N)

M = mass of the object (Kg)

Earth - g is approximately 10N/Kg

- mass of 1kg on earth's surface $\Rightarrow$ experiences a force of 10N due to earth's gravitational field

On Moon $\Rightarrow$ same mass $\Rightarrow$ experiences a gravitational force of only 1.6N

$\rightarrow$ this is because gravitational field strength on moon is 1.6N/Kg

Relation between Mass and Weight

From equation $g = \frac{W}{m}$, we have $W = mg$

Weight or gravitational field strength W acting on an object $\Rightarrow$ directly proportional to its mass m .

Gravitational Field Strength and acceleration due to gravity

if objects were to free-fall $\Rightarrow$ under gravity without air resistance $\rightarrow$ we can find its acceleration using the equation:

$$F = ma$$

F = resultant force

M = mass (Kg)

A = acceleration (m/s^2)

Object of mass m free-falling under gravity $\Rightarrow$ without air resistance will have acceleration of $a = g = 10m/s^2$ due to its weight W

Common weighing instruments

spring balance, bathroom scale, electronic balance $\rightarrow$ all measure weight of object rather than mass. But they are calibrated to give readings in grams or kilograms.

Using these weighing instruments - objects will have diff. Mass readings at different gravitational field strengths.

Refer to moon and earth example*

This means that a weighing scale calibrated for use on earth cannot be used on the moon.

$\Rightarrow$ the weighing scale must be calibrated to the moon's gravitational field strength to give accurate mass measurements on the moon.

Scale measures the normal contact force of the feet pushing down on the scale. Normal force acts to compress the spring in the bathroom scale. When person is standing still, the normal force exerted on the spring is equal to the gravitational force of attraction (weight)

Measurement of Mass

To avoid having to calibrate weighing scales for different gravitational field strengths - mass of object can be measured using a beam balance.

Beam balance $\rightarrow$ compares the gravitational force acting on an object with that acting on standard masses. Both object and standard masses experience the same gravitational field strength - mass readings taken for a given object whether on earth or moon will be the same.

Differences Between Mass and Weight

Mass	Weight
Amount of matter	Gravitational force
Scalar quantity (only has magnitude and no unit)	Vector quantity (both magnitude and direction)
SI unit: Kg	SI unit: Newton (N)
Independent of gravitational field strength	Dependent on the gravitational field strength
Measured with a beam balance or a calibrated electronic balance	measured with a spring balance

Important Point

Explain what it is meant when an astronaut says he feels weightless:

- When the astronaut is in the satellite, the only force acting on him is the force of gravity, a non-contact force. (1) The astronaut cannot feel the force of gravity unless there are contact forces opposing it. (1) When the astronaut reports that he is weightless, he means he does not experience any external objects touching and exerting a push or pull on him (1)
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